Validation Report

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Review Artifact Ids

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Report Content

SOFTWARE REQUIREMENTS SPECIFICATION (SRS) VALIDATION REPORT
Review Document ID: `review_document_Reviewer_v1.1`
SRS Artifact ID: `software_requirement_specs_Archivist_v1.2`
Date of Validation Report: 2024-06-15
Validated Against: IEEE Std 830–2013 *Recommended Practice for Software Requirements Specifications* and ISO/IEC/IEEE 29148:2018 *Systems and Software Engineering — Life Cycle Processes — Requirements Engineering*

✅ Executive Summary

The SRS for *Archivist v1.2* is a mature, well-structured, and largely standards-compliant specification. It demonstrates **strong completeness**, **robust traceability**, and **excellent verifiability foundations**, with formal acceptance criteria, test procedures, and a comprehensive traceability matrix. The document aligns closely with IEEE 830 and ISO/IEC/IEEE 29148 in scope, organization, and rigor.

However, several **high-impact gaps**—particularly in error recovery, accessibility scope clarity, security depth, terminology definition, and external dependency risk mitigation—prevent unconditional approval. These do not invalidate the core requirements but introduce ambiguity or verification risk that must be resolved before final sign-off.

Overall Assessment:
✅ **SRS is technically sound and development-ready *in principle***
⚠️ **Requires targeted revisions to address critical and major issues before formal baselining**
➡️ **Status:** `PASS_WITH_CONDITIONS`

⚠️ Issues Found (Categorized by Severity)

Severity	Description	Section / Reference	Rationale
Critical	**Missing explicit error recovery & failure handling requirements**	§1 Completeness, §3 Correctness	The SRS mandates atomic exports and hard constraint enforcement but omits *system-level recovery behavior* for failures (e.g., data corruption during export, partial write crashes, scheduling engine exceptions). Per IEEE 830 §5.1.3 and ISO/IEC/IEEE 29148 §7.2.2, recoverability is a mandatory functional concern when data integrity or

Severity	Description	Section / Reference	Rationale
			system continuity is mission-critical. Absence blocks verification of resilience.
Major	**Ambiguous and inconsistently defined key terms** (`partial schedule` , `unscheduled classes` , `hard constraints` , `soft prioritization`)	§2 Consistency	Terms appear across UC-005, FR-006, FR-001, and FR-005 without formal definitions. This undermines consistency, test design, and stakeholder alignment. ISO/IEC/IEEE 29148 §6.2.1 requires unambiguous terminology; omission risks misinterpretation and inconsistent implementation.
Major	**Inconsistent scope treatment of accessibility**	§1 Completeness (WCAG reference)	NFR-USAB-002 cites WCAG 2.1 Level AA compliance *as a requirement*, yet Introduction states "no formal accessibility certification" is in scope. This contradiction creates ambiguity: Is conformance *required*? Is *testing* required? Is *certification* excluded—or just *third-party attestation*? Must be resolved to ensure legal/regulatory compliance and testability.
Major	**Insufficient security requirements for sensitive local data**	§1 Completeness (Security)	Admin authentication exists, but no requirements for encryption-at-rest (e.g., local schedule DB), audit logging of admin actions, or tamper-evident logs—despite handling PII-adjacent data (user logs, schedules). Violates ISO/IEC/IEEE 29148 §7.3.1 (security as non-functional requirement) and university data governance expectations.
Major	**No explicit requirements for external dependency failure modes**	§3 Correctness (External Systems)	Reliance on stable external URLs (maps, department pages) is documented, but no requirements define fallback behavior, timeout policies, graceful degradation, or offline capability. Creates unverifiable assumptions and production risk.
Minor	**Lack of glossary or centralized definitions section**	§2 Consistency	While not strictly required, a glossary would significantly improve readability and reduce ambiguity for reviewers, testers, and new developers. Recommended per IEEE 830 §5.1.2.
Minor	**Missing hardware/environment assumptions for performance targets**	§3 Correctness (Timing Targets)	Performance goals (e.g., ≤90 sec for 500 classes) lack stated baseline hardware specs (CPU/RAM/storage), OS, or concurrency assumptions—making verification environment-dependent and less repeatable.
Minor	**No versioning history or change control procedure documented**	General Comments	Hinders maintainability and auditability over time. Not mandatory per IEEE 830 but strongly recommended per ISO/IEC/IEEE 29148 Annex D (configuration management integration).

> 🔍 **Note on Structural Conformance**:
> The SRS adheres to the IEEE 830 template structure (Introduction, Overall Description, Specific Requirements, External Interfaces, Appendices). All *mandatory* sections are present. However:
>
> - **Missing optional but high-value sections** (Glossary, Version History, Assumptions & Dependencies appendix) → rated **Minor**.

> - No structural omissions of required sections → **no Critical/Major structural defects**.

📋 Action Items (Prioritized)

#	Action Item	Priority	Owner (SRS Author)	Target Resolution
1	Add formal **Error Handling & Recovery Requirements** section (e.g., §5.7), specifying behavior for: data corruption during export, scheduler crash mid-run, partial file writes, and failed external API calls. Include logging, user notification, and rollback guarantees.	**Critical**	Requirements Lead	v1.3
2	Resolve **accessibility scope ambiguity**: Either (a) remove WCAG citation if out-of-scope, OR (b) add explicit requirement stating *"System shall be tested against WCAG 2.1 Level AA success criteria for all UI components"*, and clarify certification is *not required* but conformance *is*.	**Major**	Product Owner / Compliance	v1.3
3	Define and standardize **key terms** (`partial schedule` , `unscheduled classes` , `hard constraints` , `soft prioritization`) in a new **Glossary** (§1.4) or Definitions subsection. Cross-reference all uses.	**Major**	Requirements Analyst	v1.3
4	Augment security requirements with: • `SEC-ENC-001` : "All locally stored schedule and log files shall be encrypted at rest using AES-256." • `SEC-AUD-001` : "All admin login attempts and configuration changes shall be logged with timestamp, user ID, and action."	**Major**	Security Architect	v1.3
5	Add **External Dependency Failure Requirements**, e.g.: • `EXT-URL-001` : "If external map URL returns HTTP 404/503, system shall display cached building coordinates and notify user." • `EXT-URL-002` : "All external HTTP requests shall timeout after 8 seconds."	**Major**	Integration Architect	v1.3
6	Insert **Version History** appendix listing SRS versions, dates, authors, and change summaries. Add brief **Assumptions & Dependencies** subsection in §1.3.	**Minor**	Documentation Lead	v1.3
7	Specify hardware/software baseline for all performance requirements (e.g., "Tested on Intel i7-11800H, 32GB RAM, Windows 11, SSD") and add scalability thresholds (e.g., "≤180 sec for 10,000 classes under same config").	**Minor**	Performance Engineer	v1.3

📌 Overall Validation Status

Metric	Status	Notes
Structural Conformance (IEEE 830)	✅ Compliant	All mandatory sections present; optional sections missing but not critical.
Completeness	⚠️ Partially Compliant	Gaps in recovery, security depth, and dependency failure handling prevent full compliance.

Metric	Status	Notes
Consistency	⚠️ Partially Compliant	Terminology inconsistencies introduce ambiguity; resolved via Glossary.
Correctness	✅ Compliant	Logic, algorithms, and assumptions are sound—pending clarification of environmental bounds.
Verifiability	✅ Strong Foundation	Acceptance criteria exist; enhancements needed for edge-case test data and NFR measurability.
Traceability	✅ Excellent	Matrix is comprehensive and cross-referenced; minor enhancement recommended (design traceability).
Final Determination	**PASS_WITH_CONDITIONS**	SRS is fit for development *contingent upon resolution of all Critical and Major items above*.

📣 Recommendations for Next Steps

1. **Revise SRS to v1.3**, incorporating all Critical and Major action items.
2. Conduct a **focused walkthrough** with QA and DevOps to validate revised error recovery and security requirements.
3. Perform a **traceability gap analysis** to confirm 100% coverage of WCAG, FERPA-adjacent privacy assumptions, and external dependency failure modes.
4. Submit **v1.3 for re-validation** within 10 business days. A lightweight follow-up review will verify closure of Critical/Major items only.
5. Initiate parallel **test case refinement** for newly added requirements (recovery, security, fallbacks) to maintain verification velocity.

Prepared By:
Reviewer Agent (Standards-Conformance Focus)
Aligned with IEEE 830, ISO/IEC/IEEE 29148, and industry best practices

Approved For Conditional Release:
✅ Yes — pending resolution of Critical & Major items
❌ No — not approved for baselining in current form

End of Validation Report

Overall Status

fail

Summary

Validation report for SRS software_requirement_specs_Archivist_v1.2. Status: fail.